Lecture-9
Fluid Kinematics IV:
Ideal flows and potential functions

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Learning Objectives

By the end of this lecture, you should be able to

1. Prove that any **irrotational velocity field** can be mathematically represented by a **scalar velocity potential.**
2. Show that **streamlines** and **equipotential lines** are **mutually orthogonal** at every intersection point.
3. Define an **ideal flow** based purely on its **kinematic restrictions** and construct its **governing equation**
4. Utilize the **principle of superposition** to analytically combine elementary potential flows (such as uniform flows and line sources) into more complex flow fields

Velocity potential

- Since the curl of a gradient is zero, we can introduce a **scalar potential for any irrotational flow**

$$\vec{V} = -\nabla\phi$$

- The minus sign is inserted so that the **potential decreases in the direction of flow**
- In regions of flow where the **viscous forces are negligible**, irrotationality is a valid assumption

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

2D incompressible and irrotational flow

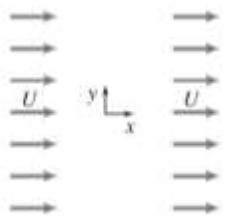
- We can define both the **velocity potential** and **streamfunction** for these flows

Definition	Always satisfies …	Satisfies Laplace equation … $\frac{\partial^2()}{\partial x^2}+\frac{\partial^2()}{\partial y^2}=\nabla^2()=0$
Stream function ψ	… incompressibility:	… only if irrotational:
$u=\frac{\partial\psi}{\partial y}\ v=-\frac{\partial\psi}{\partial x}$	$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=\frac{\partial^2\psi}{\partial x\partial y}-\frac{\partial^2\psi}{\partial y\partial x}\equiv 0$	$\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}=-\frac{\partial^2\psi}{\partial x\partial y}-\frac{\partial^2\psi}{\partial y\partial y}=0$
Velocity potential ϕ	… irrotationality:	… only if incompressible:
$u=-\frac{\partial\phi}{\partial x}\ v=-\frac{\partial\phi}{\partial y}$	$\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}=-\frac{\partial^2\phi}{\partial x\partial y}-\frac{\partial^2\phi}{\partial y\partial x}\equiv 0$	$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=-\frac{\partial^2\phi}{\partial x\partial y}-\frac{\partial^2\phi}{\partial y\partial y}=0$

- Slope of the **streamline** at any point is the negative reciprocal of the **line of constant potential**, which means that they are **orthogonal**

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Some elementary ideal flows

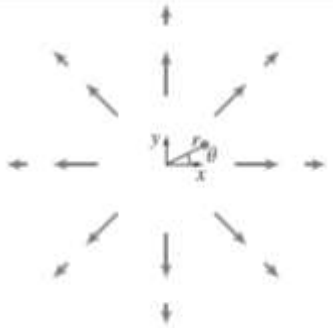


Uniform Flow (positive x direction)

$u = U \qquad \psi = Uy$

$v = 0 \qquad \phi = -Ux$

$\Gamma = 0$ around any closed curve



Source Flow (from origin)

$$V_r = \frac{q}{2\pi r} \qquad \psi = \frac{q}{2\pi}\theta$$

$$V_\theta = 0 \qquad \phi = -\frac{q}{2\pi}\ln r$$

Origin is singular point

q is volume flow rate per unit depth

$\Gamma = 0$ around any closed curve



Sink Flow (toward origin)

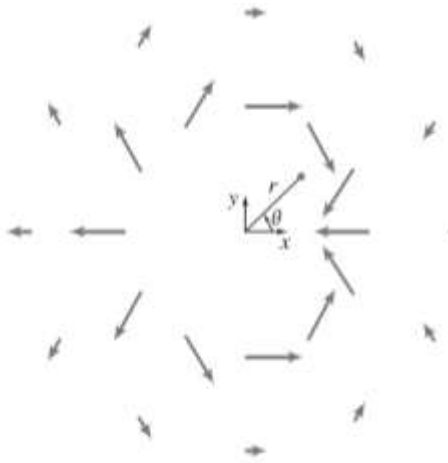
$$V_r = -\frac{q}{2\pi r} \qquad \psi = -\frac{q}{2\pi}\theta$$

$$V_\theta = 0 \qquad \phi = \frac{q}{2\pi}\ln r$$

Origin is singular point

q is volume flow rate per unit depth

$\Gamma = 0$ around any closed curve



Doublet (center at origin)

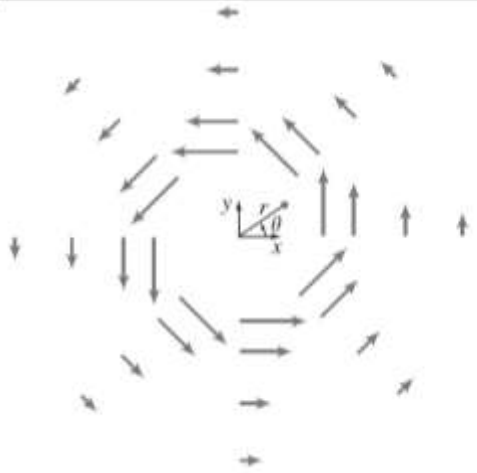
$$V_r = -\frac{\Lambda}{r^2}\cos\theta \qquad \psi = -\frac{\Lambda \sin\theta}{r}$$

$$V_\theta = -\frac{\Lambda}{r^2}\sin\theta \qquad \phi = -\frac{\Lambda\cos\theta}{r}$$

Origin is singular point

Λ is strength of the doublet

Γ = around any closed curve



Irrotational Vortex (counterclockwise, center at origin)

$$V_r = 0 \qquad \psi = -\frac{K}{2\pi}\ln r$$

$$V_\theta = \frac{K}{2\pi r} \qquad \theta = -\frac{K}{2\pi}\theta$$

Origin is singular point

K is strength of the vortex

$\Gamma = K$ around any closed curve enclosing origin

$\Gamma = 0$ around any closed curve not enclosing origin

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Superposition of ideal flows

Source and Uniform Flow (flow past a half-body)

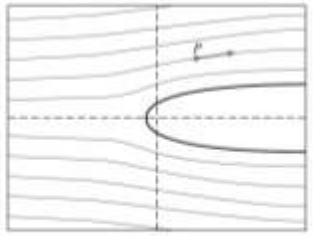


$$\psi = \psi_{so} + \psi_{uf} = \psi_1 + \psi_2 = \frac{q}{2\pi}\theta + Uy$$

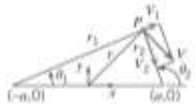
$$\psi = \frac{q}{2\pi}\theta + Ur\sin\theta$$

$$\phi = \phi_{so} + \phi_{uf} = \phi_1 + \phi_2 = -\frac{q}{2\pi}\ln r - Ux$$

$$\phi = -\frac{q}{2\pi}\ln r - Ur\cos\theta$$



Source and Sink (equal strength, separation distance on x axis = 2a)

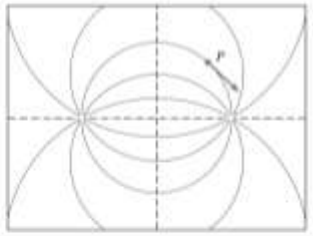


$$\psi = \psi_{so} + \psi_{si} = \psi_1 + \psi_2 = \frac{q}{2\pi}\theta_1 - \frac{q}{2\pi}\theta_2$$

$$\psi = \frac{q}{2\pi}(\theta_1 - \theta_2)$$

$$\phi = \phi_{so} + \phi_{si} = \phi_1 + \phi_2 = -\frac{q}{2\pi}\ln r_1 + \frac{q}{2\pi}\ln r_2$$

$$\phi = -\frac{q}{2\pi}\ln\frac{r_2}{r_1}$$



Source, Sink, and Uniform Flow (flow past a Rankine body)



$$\psi = \psi_{so} + \psi_{si} + \psi_{uf} = \psi_1 + \psi_2 + \psi_3$$

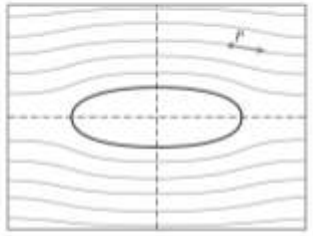
$$= \frac{q}{2\pi}\theta_1 - \frac{q}{2\pi}\theta_2 + Uy$$

$$\psi = \frac{q}{2\pi}(\theta_1 - \theta_2) + Ur\sin\theta$$

$$\phi = \phi_{so} + \phi_{si} + \phi_{uf} = \phi_1 + \phi_2 + \phi_3$$

$$= -\frac{q}{2\pi}\ln r_1 + \frac{q}{2\pi}\ln r_2 - Ux$$

$$\phi = \frac{q}{2\pi}\ln\frac{r_2}{r_1} - Ur\cos\theta$$



Vortex (clockwise) and Uniform Flow

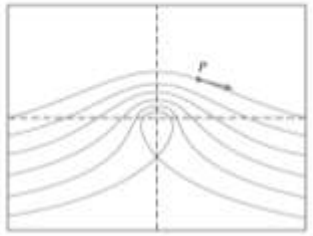


$$\psi = \psi_v + \psi_{uf} = \psi_1 + \psi_2 = \frac{K}{2\pi}\ln r + Uy$$

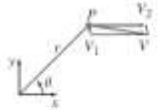
$$\psi = \frac{K}{2\pi}\ln r + Ur\sin\theta$$

$$\phi = \phi_v + \phi_{uf} = \phi_1 + \phi_2 = \frac{K}{2\pi}\theta - Ux$$

$$\phi = \frac{K}{2\pi}\theta - Ur\cos\theta$$



Doublet and Uniform Flow (flow past a cylinder)



$$\psi = \psi_d + \psi_{uf} = \psi_1 + \psi_2 = -\frac{\Lambda\sin\theta}{r} + Uy$$

$$= -\frac{\Lambda\sin\theta}{r} + Ur\sin\theta$$

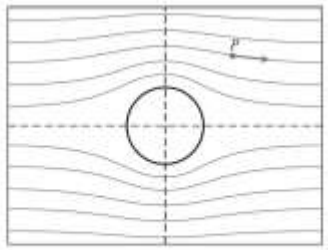
$$\psi = U\left(r - \frac{\Lambda}{Ur}\right)\sin\theta$$

$$\psi = Ur\left(1 - \frac{a^2}{r^2}\right)\sin\theta \quad a = \sqrt{\frac{\Lambda}{U}}$$

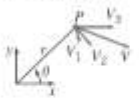
$$\phi = \phi_d + \phi_{uf} = \phi_1 + \phi_2 = -\frac{\Lambda\cos\theta}{r} - Ux$$

$$= -\frac{\Lambda\cos\theta}{r} - Ur\cos\theta$$

$$\phi = -U\left(r + \frac{\Lambda}{Ur}\right)\cos\theta = -Ur\left(1 + \frac{a^2}{r^2}\right)\cos\theta$$



Doublet, Vortex (clockwise), and Uniform Flow (flow past a cylinder with circulation)



$$\psi = \psi_d + \psi_v + \psi_{uf} = \psi_1 + \psi_2 + \psi_3$$

$$= -\frac{\Lambda\sin\theta}{r} + \frac{K}{2\pi}\ln r + Uy$$

$$\psi = -\frac{\Lambda\sin\theta}{r} + \frac{K}{2\pi}\ln r + Ur\sin\theta$$

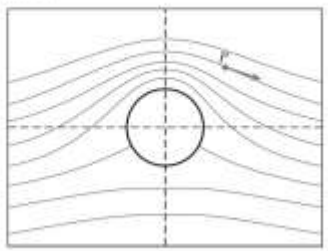
$$\psi = Ur\left(1 - \frac{a^2}{r^2}\right)\sin\theta + \frac{K}{2\pi}\ln r$$

$$\phi = \phi_d + \phi_v + \phi_{uf} = \phi_1 + \phi_2 + \phi_3$$

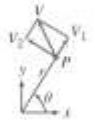
$$= -\frac{\Lambda\cos\theta}{r} + \frac{K}{2\pi}\theta - Ux$$

$$a = \sqrt{\frac{\Lambda}{U}}; K < 4\pi aU \qquad \phi = -\frac{\Lambda\cos\theta}{r} + \frac{K}{2\pi}\theta - Ur\cos\theta$$

$$\phi = -Ur\left(1 + \frac{a^2}{r^2}\right)\cos\theta + \frac{K}{2\pi}\theta$$

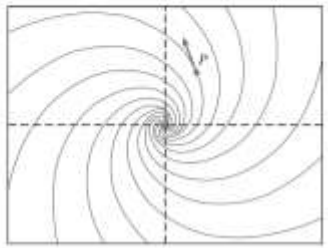


Source and Vortex (spiral vortex)



$$\psi = \psi_{so} + \psi_v = \psi_1 + \psi_2 = \frac{q}{2\pi}\theta - \frac{K}{2\pi}\ln r$$

$$\phi = \phi_{so} + \phi_v = \phi_1 + \phi_2 = \frac{q}{2\pi}\ln r - \frac{K}{2\pi}\theta$$



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Revisiting the learning objectives

1. Prove that any **irrotational velocity field** can be mathematically represented by a **scalar velocity potential.**
2. Show that **streamlines** and **equipotential lines** are **mutually orthogonal** at every intersection point.
3. Define an **ideal flow** based purely on its **kinematic restrictions** and construct its **governing equation.**
4. Utilize the **principle of superposition** to analytically combine elementary potential flows (such as uniform flows and line sources) into more complex flow fields.

